# Defence Guide · Student 2

**Student 2: Results** Health Anxiety, AI Chatbot Use and Cyberchondria Among Adults in Baghdad

> **Your role.** You carry the heaviest numbers. Your job is to make them simple: one message per slide, then the “In summary” line.

| | |
|---|---|
| **You present** | Slides 13–23 (≈ 6 minutes) |
| **You own in the thesis** | Chapter 3 Results (3.1–3.6), Tables 1–10, Figures 2–6; and Section 2.5 Statistical analysis (you are the team's statistics person) |
| **Your handover line** | “Those are our results. My colleague will now discuss what they mean.” |
| **Your teammates** | Student 1: Introduction & Methods (slides 1–12) · Student 3: Discussion & Conclusions (slides 24–33) |

## What is in this guide


1. The whole study: two-minute script, core facts, terms, “never say” list (same in all three guides)
2. Your chapter, section by section: what it says, the numbers, the sources, what to watch
3. Your key sources: each reference number in your chapter and what it found
4. Your slides, one by one: what is on screen, your script, timing and traps
5. Questions: on your part, on the whole study, and difficult ones, with model answers
6. Editing your chapter safely and in your own words
7. Your 14-day plan
8. Self-test with answers
9. Checklist for the day before

# Part 1 · The whole study (everyone must know this)

> **The committee can ask any of you about any part.** You split the presentation, not the knowledge. Learn this part so well that you can explain the whole study without slides.

## The study in two minutes (practise saying this out loud)

*"Many people now look up their symptoms online before seeing a doctor, and more and more of them ask AI chatbots like ChatGPT. For some people this searching makes them more worried, not less. That is called cyberchondria, and it is closely linked to health anxiety. No one had studied this in Baghdad, and AI chatbots had hardly been studied next to search engines. We ran an online survey of 300 adults in Baghdad between 17 July 2026 and 11 September 2026, using two established scales: the SHAI-14 for health anxiety and the CSS-12 for cyberchondria. We found that AI chatbots are now the main source of health information: 78.0% ticked them, more than search engines (56.3%). About one in five (19%) had elevated health anxiety. Health anxiety and cyberchondria were moderately linked, r = 0.47, and this held after adjusting for everything else. The link ran mainly through excessive searching and post-search distress, which fits a vicious cycle, although one survey cannot prove direction. We found no reliable evidence that AI itself adds to cyberchondria beyond health anxiety. And searching shapes the consultation: about a third delay care because they feel reassured, and about a third never tell their doctor what they searched. So our main message for clinicians is simple: ask patients, without judgement, what they searched or asked an AI."*

## Core facts (memorise; examiners test these)

| Topic | Fact |
|---|---|
| **Design** | Analytical cross-sectional online survey; STROBE and CHERRIES checklists |
| **Who and when** | **300** adults, Baghdad, 17 July 2026 – 11 September 2026; convenience + snowball sampling |
| **Sample size** | **194** needed for r = 0.20 (α 0.05, 80% power); margin of error ±5.7% |
| **Sample profile** | 62.3% aged 20–29 · 56.3% female · **44.0%** medical students · 63.3% health background · 21.7% chronic condition · 70.7% answered in Arabic |
| **Tools used** | AI chatbots **78.0%** ticked (88.7% incl. named) · search engines 56.3% · medical platforms 18.3% · social media 10.7% · ChatGPT 59.3% · AI without any search engine 41.0% |
| **Motives** | convenience 80.0% · "do I need a doctor?" 57.7% · instant reassurance 51.0% |
| **Health anxiety** | SHAI mean 12.3 ± 5.8 (0–42) · **19%** elevated (≥ 18; 95% CI 15.0–23.8%) · 0.7% at ≥ 29 |
| **Cyberchondria** | CSS mean 28.9 ± 8.4 (12–60) · excessiveness highest, compulsion lowest |
| **Reliability** | α 0.81 (SHAI), 0.86 (CSS) (Cronbach α ≥ 0.8 = good) |
| **Arabic vs English** | no difference in SHAI or CSS scores (Mann–Whitney p 0.86 and 0.65) |
| **Main result** | **r = 0.47** (95% CI 0.38–0.55); explains 22% of the variance · partial r = 0.48 |
| **By facet** | excessiveness 0.46 · distress 0.45 · compulsion 0.25 · reassurance 0.24 |
| **Regression** | $R^2$ 0.030 (demographics) → 0.133 (+ digital use) → 0.317 (+ health anxiety, $\Delta R^2$ 18.4%); β = 0.48 |
| **Who is anxious** | AOR **3.12** per SD of cyberchondria · 3.83 reassurance motive · 0.61 per older age group · AUC 0.83 |
| **Pathways** | via excessive searching 0.16 (Sobel z 5.14) · via distress 0.22 (direct → 0.01) · reverse 0.15 |
| **Behaviour** | delay care when reassured **34.7%** · ignore symptom 30.0% · several doctors 25.3% · urgent care 22.0% · self-medication 21.3% |
| **Doctor relationship** | never tell the doctor **31.7%** · always tell 27.0% · trust strained 34.7% · higher cyberchondria → more disclosure (AOR 0.57, p 0.001) |
| **AI effect** | +5.3 CSS points (inclusive) but +1.5, p = 0.21 (strict) → no reliable independent effect · 32 named-not-ticked · 34 non-users |

| | |
|---|---|
| **Robustness** | every sensitivity analysis left the main result unchanged (r 0.47–0.48) |

## Terms you must be able to explain in one sentence

| Term | Plain explanation |
|---|---|
| Health anxiety | Persistent worry about having or developing a serious illness, kept going by misreading normal body sensations and by reassurance seeking. |
| Illness anxiety disorder | The DSM-5 diagnosis at the severe end of health anxiety. We did **not** diagnose it; we measured symptoms. |
| Cyberchondria | Excessive or repeated online health searching that increases rather than relieves distress. |
| SHAI-14 | 14 items, each scored 0–3 (total 0–42); higher = more health anxiety. ≥ 18 = elevated (screening threshold). |
| CSS-12 | 12 items scored 1–5 (total 12–60) with four 3-item facets: excessiveness, compulsion, distress, reassurance seeking. |
| Cross-sectional | Everything measured once, at one time point: shows associations, not cause and effect. |
| Convenience / snowball sampling | People who were easy to reach, who then shared the link with others. Quick, but not representative. |
| Mean ± SD | The average score and how spread out the scores are. |
| Median (IQR) | The middle score and the range of the middle half; used when data are skewed. |
| p-value | How surprising the result would be if there were truly no effect; $p < 0.05$ is called significant. |
| 95% CI | The range in which the true value probably lies; narrower = more precise. |
| Correlation r | How closely two scores move together (0 = none, 1 = perfect). 0.47 is moderate. |
| Partial correlation | The same link after removing the effect of other variables. |
| Mann–Whitney / Kruskal–Wallis | Non-parametric tests comparing two / three or more groups, used because scores were not normally distributed. |
| $\chi^2$ test | Tests whether two categorical variables are related (e.g., social-media use and elevated health anxiety). |
| Sobel test | Tests whether an indirect (mediated) effect a × b is different from zero. |
| Linear regression, B and β | Predicts a score from several factors at once. B = change in CSS points per unit; β = the same on a standardised scale (to compare factors). |
| $R^2$ and $\Delta R^2$ | Share of variation explained by the model; $\Delta R^2$ = extra share explained by adding a block of variables. |
| Logistic regression, AOR | Predicts a yes/no outcome. Adjusted odds ratio: how many times the odds change, holding other factors constant. |
| AUC | How well a model separates two groups (0.5 = chance, 1 = perfect). |
| Mediation | Splitting a link into a part that runs through a third variable (indirect) and the rest (direct). Statistical, not causal. |
| Hosmer–Lemeshow test | Checks the fit of a logistic model; $p > 0.05$ = good fit. |
| Cronbach α | Internal consistency: do the items of a scale hang together? ≥ 0.7 is acceptable, ≥ 0.8 good. |
| Cramér's V | Strength of a $\chi^2$ association: 0.1 small, 0.3 medium, 0.5 large. |
| VIF | Checks that regression predictors are not near-duplicates; below 5 is fine. |
| Sensitivity analysis | Re-running the main analysis in a different way to see if the result holds. |

**Never say** (overclaims the committee will catch):

- "Health anxiety *causes* cyberchondria" → say "is associated with", "is compatible with a cycle".
- "19% *have* health anxiety disorder" → "19% scored in the elevated range on a screening tool".
- "AI makes people anxious" → "we found no reliable independent effect of AI use".
- "Our results represent Baghdad" → "a young, online, convenience sample; associations were consistent across subgroups".
- "The Arabic version is validated" → "the team wrote the Arabic wording; reliability was good and Arabic and English scores did not differ, but there was no formal back-translation".

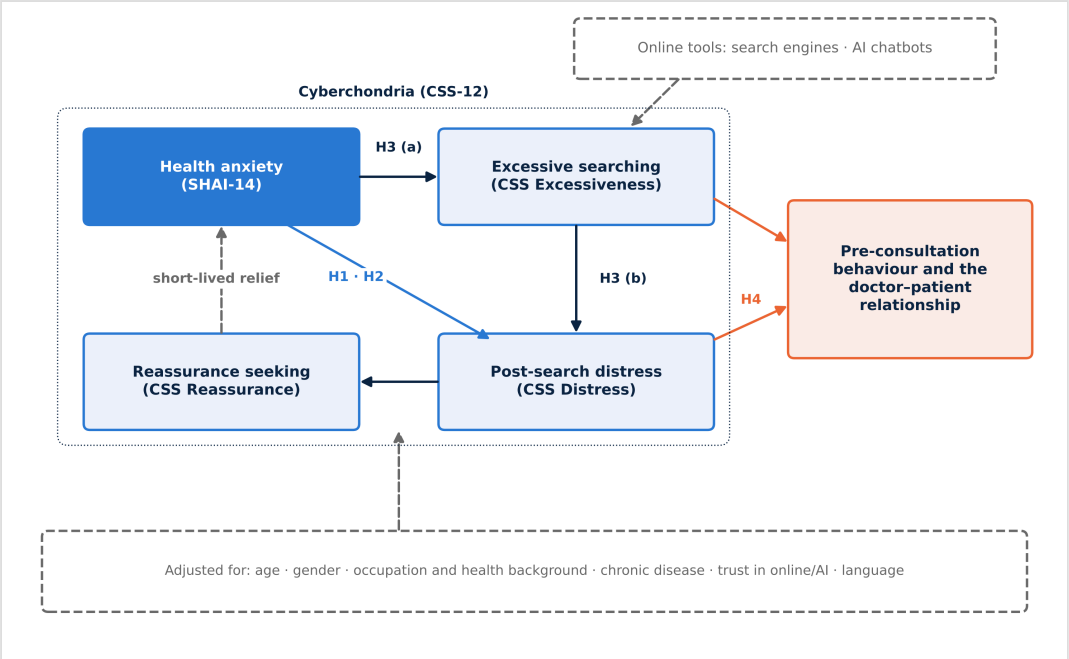



Figure 1 of the thesis: the conceptual framework. Learn to explain every arrow.

# Part 2 · Your chapter, section by section

For each section: its **purpose**, the **key points and numbers** (with the reference numbers used in the thesis), and what to **watch**.

### 3.1 Participants and online health-information seeking (Tables 1–2, Figures 2–3)

*Who answered, how they search, and what they do after searching.*

- 300 adults, no missing data; 62.3% aged 20–29; 56.3% female; medical students 44.0%; health background 63.3%; chronic condition 21.7%; Arabic 70.7%.
- AI ticked 78.0% (88.7% incl. named) vs search engines 56.3%; AI without any search engine 41.0%; ChatGPT 59.3%. AI use 96% at 18–19 y vs 72% at ≥ 40 y.
- Motives: convenience 80.0%, "do I need a doctor?" 57.7%, reassurance 51.0%.
- After searching: delay/skip care when reassured 34.7%; ignore 30.0%; several doctors 25.3%; urgent care 22.0%. Always tell the doctor 27.0%; never (fear of judgement) 31.7%; relationship strained 34.7%.

Watch: Message: a young, digitally active sample whose first "doctor" is often an AI; searching changes the consultation.

### 3.2 Levels of health anxiety and cyberchondria (Tables 3–4)

- SHAI 12.3 ± 5.8; elevated 19% (CI 15.0–23.8%); Table 4: 32.7% at ≥ 15, 19.0% at ≥ 18, 7.3% at ≥ 22, 0.7% at ≥ 29.
- CSS 28.9 ± 8.4; excessiveness highest, compulsion lowest.
- Cronbach α 0.81 (SHAI) and 0.86 (CSS): good. Arabic and English responders did not differ (Mann–Whitney p 0.86 and 0.65).

Watch: Why several cut-offs? Because prevalence depends on the threshold; reporting all four is honest.

### 3.3 Main association (Table 5, Figure 4)

*The central result (H1, H2).*

- Pearson r = 0.47 (95% CI 0.38–0.55), Spearman ρ = 0.47, p < 0.001; explains 22%; partial r = 0.48.
- Facets: strongest with excessiveness 0.46 and distress 0.45, weakest with compulsion 0.25 and reassurance 0.24.

Watch: H1 and H2 are answered here.

### 3.4 Factors associated (Tables 6–7, Figure 5)

*Mann–Whitney / Kruskal–Wallis and $\chi^2$ tests ($p < 0.05$).*

- Health anxiety higher with a chronic condition (p 0.004), in reassurance seekers (p < 0.001), social-media users (p 0.001) and AI users (p 0.008); differs by age (p 0.008), education (p 0.002) and occupation (p 0.005); gender and language no difference.
- Cyberchondria higher in AI users (p < 0.001), highest in AI + search engine users (p < 0.001), and in social-media users (p 0.021).
- Elevated health anxiety → 52.6% vs 23.0% high cyberchondria (OR 3.71) (Table 7).

Watch: Many tests at p < 0.05: a few weaker results could be chance (we say so in the limitations).

### 3.5 Independent predictors (Tables 8–9, Figure 6)

- Hierarchical linear regression: $R^2$ 0.030 → 0.133 → 0.317; health anxiety adds 18.4%; B = 0.68, β = 0.48, p < 0.001. Ages 30–39 (B 6.3) and ≥ 40 (B 4.0) also higher than 20–29.
- AI: +5.3 inclusive vs +1.5, p 0.21 strict → no reliable independent effect.
- Binary logistic (57 events): AOR 3.12 cyberchondria per SD, 3.83 reassurance motive, 0.61 age; Nagelkerke $R^2$ 0.31; Hosmer–Lemeshow p 0.50 (good fit); AUC 0.83; ≈ 5 events per variable → estimates may be overestimated.
- Sensitivity (one sentence): r stayed 0.47–0.48 in every check.

Watch: You will be asked about AOR, AUC and Hosmer–Lemeshow.

### 3.6 Behaviour and pathways (Tables 10–11, Figures 7–8)

- Higher CSS → urgent visits (AOR 1.51, p 0.013), less ignoring (0.65, p 0.009), strained relationship (1.35, p 0.041) and more disclosure (never-disclose AOR 0.57, p 0.001).
- Higher SHAI → concealment (never-disclose AOR 1.47, p 0.017). Trust → fewer multiple doctors (AOR 0.59, p 0.004).
- Mediation (Sobel test): via excessive searching 0.16 (z 5.14, p < 0.001; about a third); via distress 0.22 (z 5.98), direct 0.01 (ns); reverse 0.15 (z 4.66).

Watch: Always end with: “compatible with a cycle, not proof of direction”.

## How to read your figures

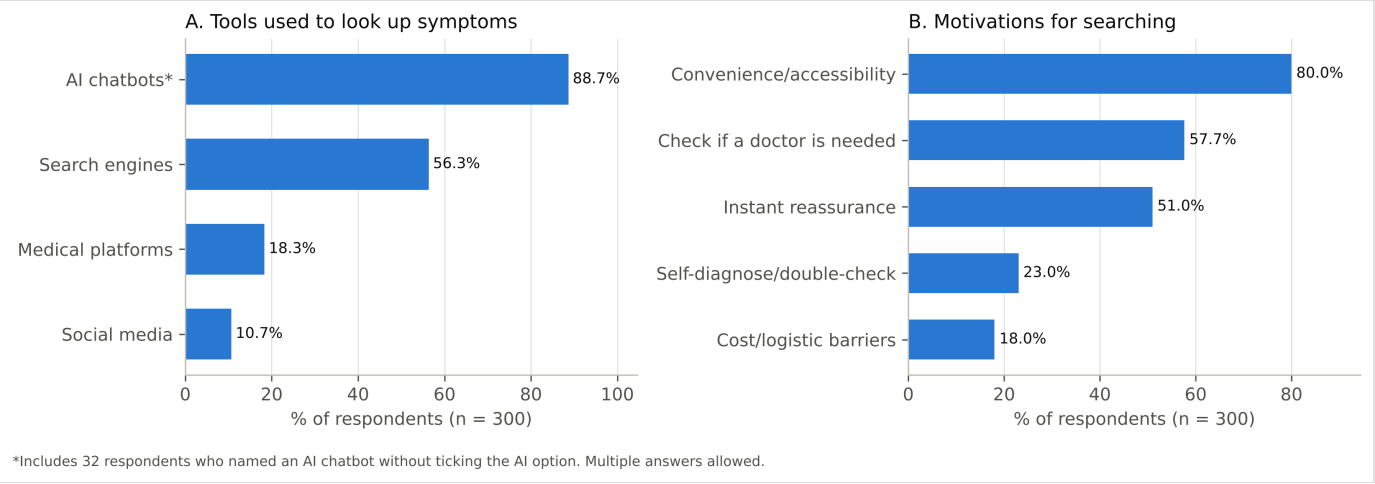



Figure 2: tools and motives. Read: bars = % of participants; multiple answers allowed, so they add to more than 100%.

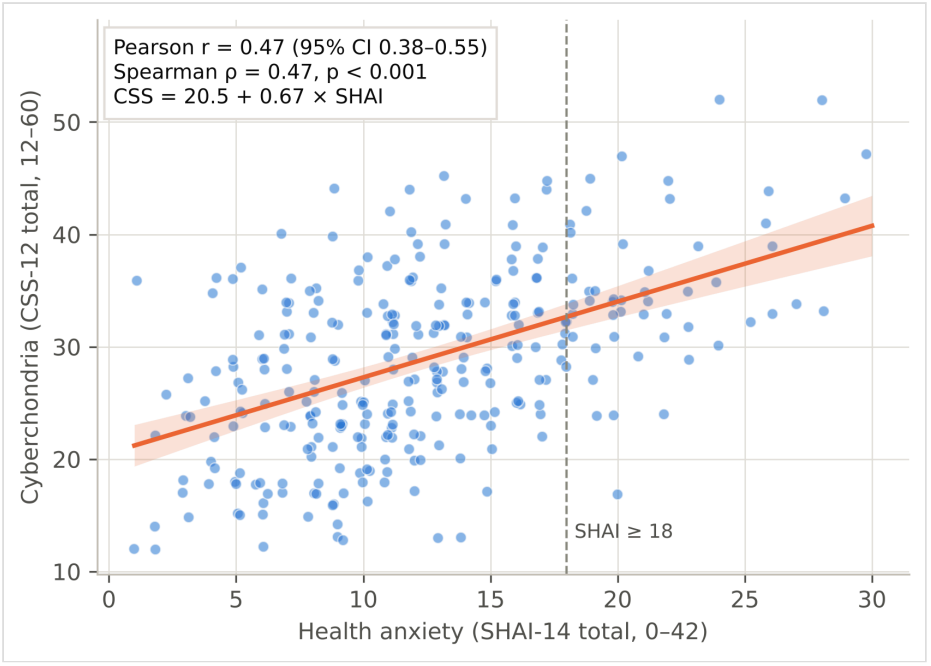



Figure 4: each dot is a participant; the line is the best fit, the band its 95% CI. The upward slope is r = 0.47.

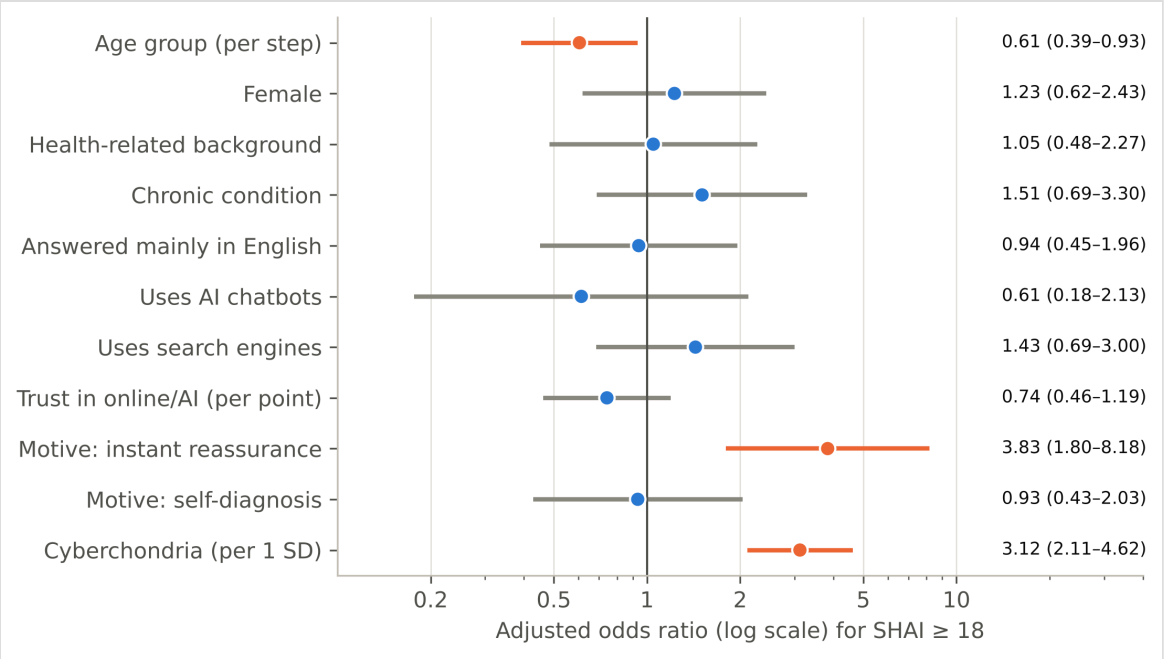



Figure 6: each dot is an AOR with its 95% CI; the vertical line at 1 = no effect; lines that cross 1 are not significant.

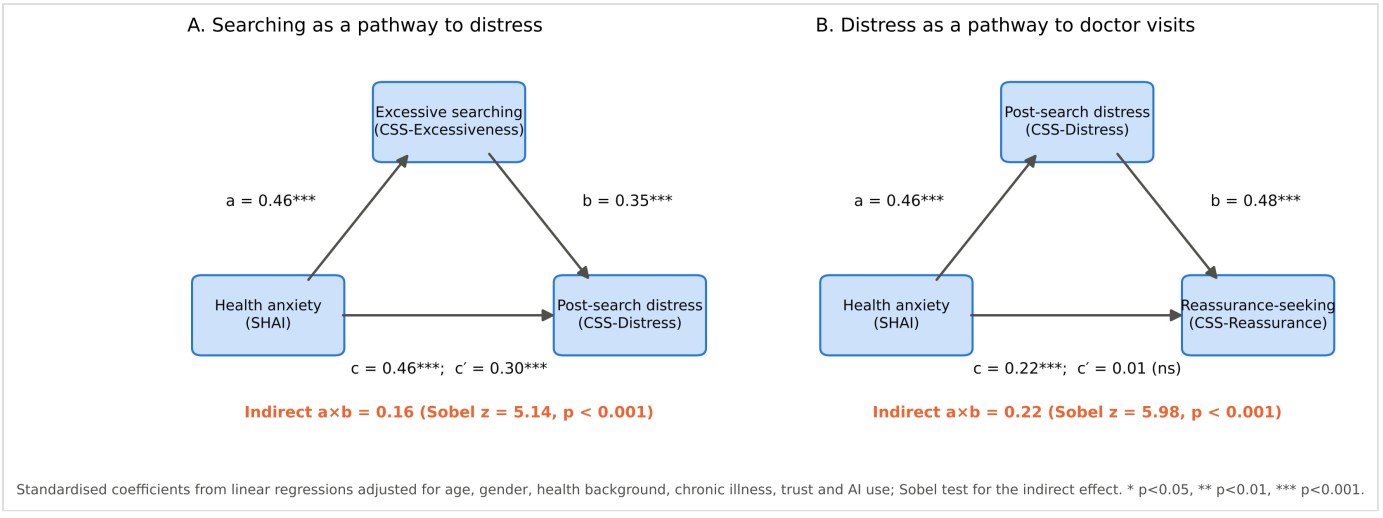



Figure 7: a = predictor → mediator, b = mediator → outcome, c′ = direct effect; the indirect effect is a × b, tested with the Sobel test.

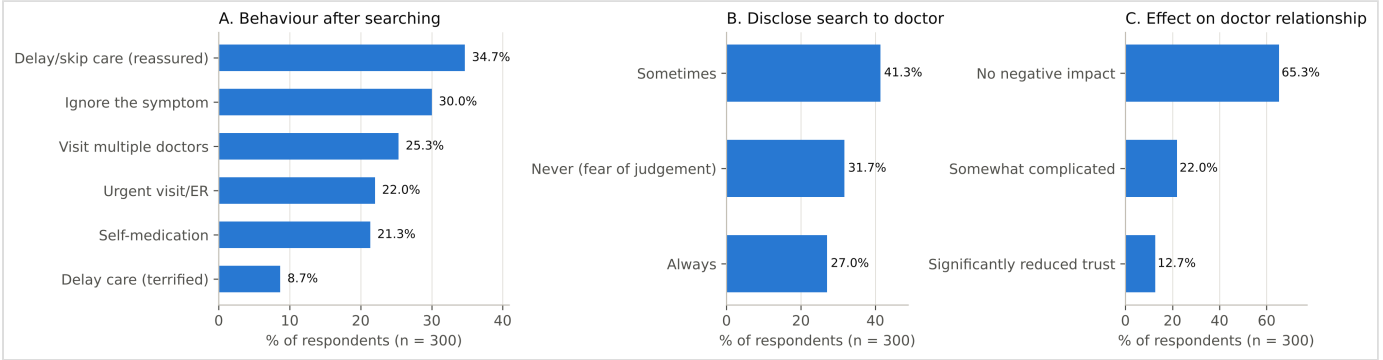



Figure 3: behaviour after searching, disclosure, and effect on the relationship.

## Part 3 · Your key sources

Every reference cited in your chapter, with its number in the thesis. Know the bold ones well: examiners often ask “who found that?”.

| No. | Reference (as listed in Chapter 6) |
|---|---|
| [8] | McElroy E, Kearney M, Touhey J, Evans J, Cooke Y, Shevlin M. The CSS-12: development and validation of a short-form version of the cyberchondria severity scale. Cyberpsychol Behav Soc Netw. 2019;22(5):330-5. doi:10.1089/cyber.2018… |
| [9] | Hallit S, Rogoza R, Abi Semaan C, Azzi V, Sawma T, Obeid S. Validation of the Arabic version of the cyberchondria severity scale 12 items (CSS-12-Ar) among a sample of Lebanese adults. BMC Psychiatry. 2023;23(1):618. doi:10.1186/s… |
| [21] | von Elm E, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. Lancet. 2007;370… |
| [22] | Eysenbach G. Improving the quality of Web surveys: the Checklist for Reporting Results of Internet E-Surveys (CHERRIES). J Med Internet Res. 2004;6(3):e34. doi:10.2196/jmir.6.3.e34 |
| [23] | Serdar CC, Cihan M, Yücel D, Serdar MA. Sample size, power and effect size revisited: simplified and practical approaches in pre-clinical, clinical and laboratory studies. Biochem Med (Zagreb). 2021;31(1):010502. doi:10.11613/BM.2… |
| [24] | Salkovskis PM, Rimes KA, Warwick HMC, Clark DM. The Health Anxiety Inventory: development and validation of scales for the measurement of health anxiety and hypochondriasis. Psychol Med. 2002;32(5):843-53. doi:10.1017/S00332917020… |
| [25] | Arnáez S, García-Soriano G, López-Santiago J, Belloch A. The Spanish validation of the Short Health Anxiety Inventory: psychometric properties and clinical utility. Int J Clin Health Psychol. 2019;19(3):251-60. doi:10.1016/j.ijchp… |
| [26] | Österman S, Axelsson E, Lindefors N, Hedman-Lagerlöf E, Hedman-Lagerlöf M, Kern D, et al. The 14-item short health anxiety inventory (SHAI-14) used as a screening tool: appropriate interpretation and diagnostic accuracy of the Swe… |
| [27] | Cruchinho P, López-Franco MD, Capelas ML, Almeida S, Bennett PM, Miranda da Silva M, et al. Translation, cross-cultural adaptation, and validation of measurement instruments: a practical guideline for novice researchers. J Multidi… |
| [28] | Alshayea AK. Latent structure, measurement invariance, and reliability of an Arabic version of the Short Health Anxiety Inventory. J Exp Psychopathol. 2020;11(2). doi:10.1177/2043808720912629 |
| [29] | Baron RM, Kenny DA. The moderator–mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations. J Pers Soc Psychol. 1986;51(6):1173–82. doi:10.1037/0022-3514.51.6.1173 |
| [30] | Sobel ME. Asymptotic confidence intervals for indirect effects in structural equation models. Sociol Methodol. 1982;13:290–312. doi:10.2307/270723 |

## Part 4 · Your slides, one by one

The script below is the speaker-notes text of your slides in the presentation. Practise until you can say it in your own words without reading.

### Slide 13 · Results

*Thank you. I will present the results in six steps: who took part, how they search, the levels of health anxiety and cyberchondria, the main association, the predictors and pathways, and finally behaviour and AI chatbots.*

### Slide 14 · Participants

*Opening: Table 1 shows our 300 participants. Main finding: the sample is young, 62.3% aged 20 to 29, slightly more women at 56.3%, and 44.0% are medical students; 63.3% have a health-related background, and 21.7% reported a chronic condition. 70.7% answered mainly in Arabic. Closing: so this is a young, educated, digitally active sample, and that matters when we interpret percentages.*

### Slide 15 · How People Search

*Opening: Figure 2 shows the tools people use and why. Main finding: AI chatbots are now the main gateway. 78.0% ticked AI chatbots, 88.7% if we include those who named one, compared with 56.3% for search engines. 41.0% use AI with no search engine at all, and ChatGPT alone is used by 59.3%. The top motives are convenience at 80.0%, checking whether a doctor is needed at 57.7%, and instant reassurance at 51.0%. Closing: in summary, for this sample, the first doctor is often an AI chatbot.*

### Slide 16 · Levels & Reliability

*Opening: Table 3 shows the scores and reliability. Main finding: mean SHAI was 12.3 out of 42 and 19% scored 18 or more, our threshold for elevated health anxiety, with a 95% confidence interval of 15.0–23.8%. Mean CSS was 28.9 out of 60. Excessiveness was the highest facet and compulsion the lowest: people search a lot, but it rarely disrupts their lives. Both scales were reliable, alpha 0.81 and 0.86, and scores did not differ between Arabic and English responders. Closing: our tools measured well in this population.*

### Slide 17 · Health Anxiety ↔ Cyberchondria

*Opening: this is our main result, Figure 4. Main finding: health anxiety and cyberchondria rise together, r = 0.47, 95% CI 0.38–0.55, so health anxiety explains about 22% of the variation in cyberchondria. After adjusting for every other variable, the partial correlation is still 0.48. The link is strongest for excessive searching, 0.46, and distress, 0.45, and weaker for compulsion and reassurance. Closing: in summary, hypotheses 1 and 2 are supported.*

### Slide 18 · What Predicts Cyberchondria?

*Opening: Table 8 is the hierarchical regression for cyberchondria. Main finding: demographics alone explain only about 3% of cyberchondria. Adding digital-use variables brings it to 13%. Adding health anxiety adds another 18.4%, to about 32%. Health anxiety is the strongest predictor, standardised beta 0.48: each extra SHAI point adds about 0.68 CSS points. Closing: who you are matters little; how anxious you are matters most.*

### Slide 19 · Item by Item: What Drives Searching?

*Opening: this supplementary figure breaks the association down item by item, all 168 pairs. Main finding: the strongest links involve fear of serious illness and interpreting body sensations; for example fear of illness with panicking when reading about a rare condition. Simply noticing aches and pains was almost unrelated. Closing: what drives searching is fear and interpretation, not just noticing symptoms, exactly what the cognitive model predicts.*

### Slide 20 · Who Has Elevated Health Anxiety?

*Opening: Figure 6 shows who has elevated health anxiety, in a multivariable logistic model. Main finding: three factors are independent. Each standard deviation of cyberchondria triples the odds, adjusted odds ratio 3.12. Searching for instant reassurance almost quadruples them, 3.83. Older age is protective, 0.61 per age step. The model fits well, Hosmer–Lemeshow p 0.50, and discriminates well, AUC 0.83. Closing: the person most at risk is young and searches to feel reassured.*

### Slide 21 · Pathways: Is There a Cycle?

*Opening: Figure 7 shows the pathways. Main finding: in model A, part of the link from health anxiety to post-search distress runs through excessive searching, indirect effect 0.16, Sobel z 5.14, p below 0.001. In model B, post-search distress carries almost all of the link from health anxiety to reassurance-seeking visits: the direct effect drops to 0.01. But the reverse direction also fits, 0.15. Closing: our data are compatible with the cycle, but one survey at one time point cannot prove direction. Hypothesis 3 is supported statistically, not causally.*

### Slide 22 · Behaviour Before the Consultation

*Opening: Figure 3 shows what people do after searching. Main finding: the most common response is to delay or skip care because they feel reassured, 34.7%. Others escalate: urgent visits 22.0%, multiple doctors 25.3%. Only 27.0% always tell their doctor about their searches, and 31.7% never do, for fear of being judged. Higher cyberchondria was linked with more disclosure, odds ratio 0.57, and also with urgent visits and a strained relationship. 34.7% say searching complicated or reduced trust in their doctor. Closing: searching changes the consultation before it begins.*

### Slide 23 · AI Chatbots: An Independent Effect?

*Opening: finally, did AI chatbots themselves matter? Main finding: AI users scored 5.3 points higher on cyberchondria after adjustment, mainly excessiveness and reassurance. But 32 people named a chatbot without ticking the AI option. When we use the strict definition, only people who ticked it, the effect disappears: B = 1.5, p = 0.21. And only 34 people were true non-users. Closing: no reliable independent effect of AI. Absence of evidence, not evidence of absence. I now hand over to my colleague for the discussion.*

**Delivery for your part:** ≈ 6 minutes; about 25–35 seconds per content slide and 10 seconds per divider. Point at the number you are saying. Pause after each key number. End with: “Those are our results. My colleague will now discuss what they mean.”

# Part 5 · Questions and model answers

## A. Questions on your part

**Q: Why non-parametric tests?**

Shapiro–Wilk showed the scores were not normally distributed; Mann–Whitney and Kruskal–Wallis do not assume normality.

**Q: Explain r = 0.47 to a non-statistician.**

People with more health anxiety clearly tend to have more cyberchondria; anxiety accounts for about a fifth (22%) of the differences.

**Q: Why both Pearson and Spearman?**

Pearson for the linear correlation, Spearman as a check that does not depend on normality; both were 0.47.

**Q: What does partial r = 0.48 tell us?**

The link is not explained by age, gender, background, chronic disease, trust or AI use: it survives adjustment (H2).

**Q: Why hierarchical regression?**

To see what health anxiety adds after demographics and digital use: 18.4% extra variance, by far the biggest block.

**Q: What do B and β mean in Table 8?**

B = 0.68: each SHAI point adds about 0.7 CSS points. β = 0.48: the standardised effect, used to compare predictors.

**Q: What is an adjusted odds ratio of 3.12?**

For each standard deviation higher cyberchondria, the odds of elevated health anxiety are about three times higher, holding other factors constant.

**Q: Why is the reassurance-motive AOR so wide?**

Only 57 events for 11 predictors (≈ 5 per variable), so estimates are imprecise and probably optimistic; we interpret them cautiously.

**Q: Does the mediation prove a vicious cycle?**

No. One time point, the reverse order fits too, and the facets come from one questionnaire. It is compatible with a cycle; only longitudinal data can show direction.

**Q: Why does the AI effect disappear?**

It depends on 32 people who named a chatbot without ticking it; they had the highest scores. Under the strict definition p = 0.21. Only 34 non-users made a small comparison group.

**Q: Chronic illness was not significant in the logistic model. Why?**

Dichotomising SHAI at 18 loses information and power. On the continuous score chronic illness is significant: p = 0.004 unadjusted and p = 0.029 adjusted (B +1.7).

**Q: Why is high cyberchondria defined as > 33?**

There is no validated clinical cut-off for the CSS-12, so we used the upper tertile of our sample, as is common.

**Q: Why is the prevalence so dependent on the cut-off (Table 4)?**

Because most scores cluster just below the thresholds; that is why we report several cut-offs and say “elevated”, not “disorder”.

**Q: Which software did you use?**

IBM SPSS Statistics version 26; the syntax file reproduces every table and figure.

**Q: Why didn't you correct for multiple testing?**

We used the standard $p < 0.05$, as most graduation theses do, and we say in the limitations that with many tests some weaker results may be chance. Our main results are $p < 0.001$.

**Q: What is the AUC?**

The area under the ROC curve: how well the model separates elevated from non-elevated anxiety. 0.83 is good.

**Q: What does the Hosmer–Lemeshow test tell you?**

Whether the logistic model fits the data. We want it non-significant: $p = 0.50$, so the fit is good.

**Q: How did you test mediation?**

Three linear regressions per model (paths a, b, c and c′) and the Sobel test for the indirect effect $a \times b$.

**Q: What does Sobel z = 5.14 mean?**

The indirect effect is about five standard errors above zero, so $p < 0.001$: part of the link runs through excessive searching, statistically.

**Q: Why Mann–Whitney and not the t-test?**

Shapiro–Wilk showed the scores were not normal ($p < 0.01$), so we used non-parametric tests that compare ranks.

## B. Questions anyone can be asked (know the answer even if it is not your part)

**Q: Summarise your study in one minute.**

Use the two-minute script in Part 1, shortened: design, sample, two scales, $r = 0.47$, AI result, behaviour, clinical message.

**Q: What is your main finding?**

Health anxiety and cyberchondria are moderately associated ($r = 0.47$), independent of other factors.

**Q: What did you find about AI chatbots?**

They are the main source (78.0% ticked), but there was no reliable independent effect on cyberchondria beyond health anxiety.

**Q: What percentage had elevated health anxiety?**

19% at SHAI ≥ 18, a screening threshold, not a diagnosis.

**Q: What is your biggest limitation?**

The cross-sectional design: no direction of cause.

**Q: What is the clinical message?**

Ask patients, without judgement, what they searched or asked an AI.

**Q: Why did you split the presentation this way?**

By chapter, but each of us studied the whole thesis.

**Q: Were your hypotheses written before you collected the data?**

Answer truthfully. A safe true answer: they follow from our objectives and previous research, and the questionnaire was designed to measure these things; we tested all of them and report every result, including those not supported.

Watch out: Never claim they were pre-registered.

**Q: Who did the statistical analysis?**

Answer honestly according to what you did. If AI tools helped with code or drafting, say so in line with your college policy and the declaration in the thesis.

## C. Difficult or hostile questions

### Q: “Your p-values are tiny because your sample is large.”

The effect sizes are meaningful too: r = 0.47 is moderate and $\Delta R^2$ 18.4% is a medium effect.

### Q: “You ran so many tests; something will be significant.”

Yes, that is possible for the weaker results, and we say so in the limitations. The main findings (r = 0.47, health anxiety in the regression, the mediation paths) have $p < 0.001$ and would survive any correction.

### Q: “The mediators are subscales of the same scale. Isn't that circular?”

Partly, yes: shared method variance inflates the paths. That is why we call it a statistical decomposition, not a mechanism.

## D. Limitations and what was done about each (all students)

| Limitation | What we did / how to answer |
|---|---|
| Cross-sectional design | Reverse-direction models; wording “compatible with”; longitudinal studies recommended |
| Convenience/snowball sample | Percentages not generalised; associations consistent across subgroups |
| Arabic wording (no back-translation) | Good reliability; Arabic and English scores did not differ |
| Single-item behaviour measures | Reported descriptively; interpreted cautiously |
| Many tests | Stated in the limitations; main results $p < 0.001$ |
| Small subgroups (34 non-users) | Acknowledged; AI effect depends on the definition |
| Self-report and careless responding | Anonymous; straight-line responders excluded in a sensitivity analysis (no change) |

> **How to answer any question:** pause → repeat the question briefly → answer in two or three sentences → stop. If you do not know: “We did not examine that directly; based on our data we would expect…, and it is a good question for future research.” Never invent a number. If a teammate struggles: “If I may add…”.

# Part 6 · Editing your chapter

> **The Word file is the master copy.** Edit only the .docx, in one shared OneDrive / Word Online file with **Review → Track Changes** on. Keep an untouched copy (Thesis_ORIGINAL.docx). Avoid Google Docs: it breaks the automatic numbering.

## For your chapter

- Results is mostly numbers: **do not change any number**, p, CI, test name or table reference.
- You may reword the connecting sentences in your own words; keep one main message per paragraph.
- Never type table or figure numbers by hand in captions; in text, keep "Table n / Figure n" exactly.
- If a number or sentence looks wrong, add a comment and ask; do not "fix" statistics yourself.
- Read 2.5 Statistical analysis until you can explain each method in one sentence; you will be asked.

## Never touch

- Numbers, %, p, q, CI, r, OR, β, n
- Reference numbers like [18] (move them with their sentence)
- The reference list
- Table/figure captions, table of contents and lists (automatic fields)
- Heading numbers and styles
- Times New Roman 14 and page numbering

## The rewrite method (paragraph by paragraph)

1. Read until you understand it; look up anything unclear.
2. Close it and explain it out loud.
3. Write your version from that explanation.
4. Compare: same facts, numbers and [n]; nothing claimed more strongly.
5. Read aloud; change any phrase you would never say.

## After editing

1. Ctrl + A, then F9, to update the table of contents, the lists and the captions.
2. Search for "[" to find placeholders left.
3. Swap chapters with a teammate for checking.
4. Export to PDF and look at every page.

> **About AI use, honestly.** Check your college's AI policy with your supervisor and declare AI assistance if required. A sentence you can adapt: *"The statistical analysis code and an initial draft of this report were prepared with the assistance of an AI tool (Claude, Anthropic). All analyses were checked, and the text was reviewed and rewritten by the authors, who take full responsibility for its content."* Do not use paraphrasing tools or tricks to fool detectors: they make the text worse and can count as misconduct. Real ownership (your words, your understanding, your edit history) is the best protection.

## Part 7 · Your 14-day plan

| Day | Task |
|---|---|
| 1 | Read the Abstract, 1.8 and Chapter 5. Learn the two-minute script (Part 1). |
| 2 | Read the whole thesis once, marking what you cannot explain. |
| 3 | Core facts and terms (Part 1). Self-test on core facts. |
| 4 | Your chapter, sections in Part 2 (first half). |
| 5 | Your chapter, second half; your key sources (Part 3). |
| 6 | Start rewriting your chapter (Part 6). |
| 7 | Finish rewriting. Self-test (Part 8). |
| 8 | Swap chapters with a teammate; check accuracy and voice. |
| 9 | Your slides: read the scripts (Part 4), then present them alone 3 times. |
| 10 | Questions A–D out loud (Part 5). Record yourself once. |
| 11 | Team: full rehearsal with timing and handovers. |
| 12 | Team: mock committee; each student asks the others their Part 5 questions. |
| 13 | Fix weak answers; second full rehearsal; final Word checks. |
| 14 | Day before: checklist (last page); one calm run-through; sleep. |

## Part 8 · Self-test

Cover the answers, answer out loud, then check. Repeat until all are correct.

| # | Question | Answer |
|---|---|---|
| 1 | Main correlation with 95% CI? | r = 0.47 (0.38–0.55) |
| 2 | Partial correlation? | 0.48 |
| 3 | Which two facets correlate most with health anxiety? | Excessiveness (0.46) and distress (0.45) |
| 4 | $R^2$ of the three regression blocks? | 0.030 → 0.133 → 0.317 |
| 5 | β and B for health anxiety in the final model? | β 0.48, B 0.68 |
| 6 | Three independent predictors of elevated health anxiety and their AORs? | Cyberchondria 3.12, reassurance motive 3.83, age 0.61 |
| 7 | Direct effect of HA on reassurance seeking after distress? | 0.01 (not significant) |
| 8 | AI effect inclusive vs strict? | +5.3 vs +1.5 (p 0.21) |
| 9 | Prevalence at cut-offs 15/18/22/29? | 32.7% / 19.0% / 7.3% / 0.7% |
| 10 | Reliability α of both scales? | 0.81 and 0.86 |
| 11 | AUC and Hosmer–Lemeshow p? | 0.83; p 0.50 |
| 12 | Indirect effect via excessive searching and its Sobel z? | 0.16; z 5.14 |
| 13 | Behaviours linked to higher cyberchondria? | Urgent visits, less ignoring, strained relationship, more disclosure |
| 14 | Which significance level did you use? | $p < 0.05$, two-sided, no correction for multiple tests |
| 15 | Main correlation (everyone)? | r = 0.47 |
| 16 | Elevated health anxiety (everyone)? | 19% |

| | | |
|---|---|---|
| 17 | AI chatbots ticked (everyone)? | 78.0% |
| 18 | Never tell the doctor (everyone)? | 31.7% |
| 19 | Sample size (everyone)? | 300 |

## Part 9 · Day-before checklist

| ✓ | Item |
|---|---|
| ☐ | All placeholders filled (names, supervisor, college, logo, degree, year, ethics, Google Forms question, Acknowledgment) |
| ☐ | Fields updated (Ctrl + A, F9); PDF checked page by page |
| ☐ | AI declaration added if required |
| ☐ | Presentation: names on slide 1, ethics number on slide 10; USB + PDF + email copies; clicker tested |
| ☐ | I can give the two-minute summary and my slides (13–23) without notes |
| ☐ | I have answered every question in Part 5 out loud |
| ☐ | I scored 100% on the self-test |
| ☐ | Printed thesis copy if required; arrive early; agree who stands where |

Generated from the study outputs and the presentation, so every number and slide script here matches the thesis and the deck.